// PORTFOLIO · MECHANICAL ENGINEERING

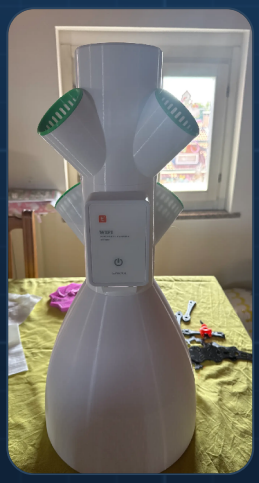
Maryan Kovba

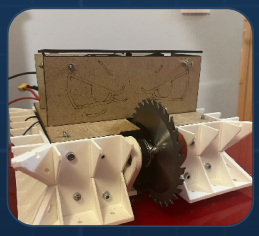
CAD Design & Rapid Prototyping

I design mechanical parts and assemblies in Fusion 360 and SolidWorks, prepare them for 3D printing, and take them all the way to a working prototype. Available for remote freelance work and internships.

kovbamaryan@gmail.com · linkedin.com/in/maryan-kovba-ba6365419 · freelancer.com/u/MaryanVK

5★	3	4
client review	built prototypes	languages







// ABOUT ME

From sketch to working prototype

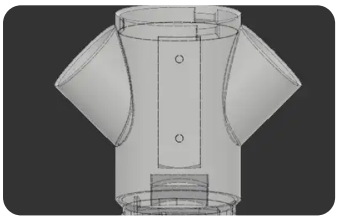
I am an Industrial Mechanical Engineering student at the University of Salamanca (Béjar campus, Spain), expected to graduate in 2028. I like building things that work in the real world: since 2025 I have been working as a freelance CAD designer for international clients, and I have built two robots with my university teams.

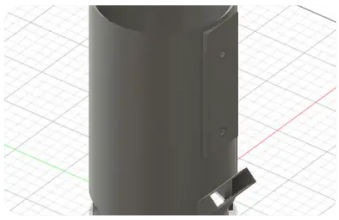
I am fast at turning an idea or a sketch into a printable, well-fitting part. When something does not work, I go back, find the problem and fix it — like the robot firmware I rewrote this summer, or the industrial printer I helped repair at a bottling plant.

SPEC SHEET

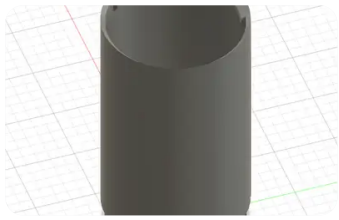
Studies	**Industrial Mechanical Eng. · USAL**
Based in	**Béjar, Spain (CET)**
Work mode	**Remote · freelance or internship**
Main tools	**Fusion 360 · SolidWorks**
Languages	**ES · UK · EN · DE**











FREELANCE · CLIENT PROJECT

Modular Hydroponic Tower

Hydroinvent Group (Italy) · Freelancer.com · 2025–2026

CHALLENGE

The client needed the main parts of a vertical hydroponic tower: large, watertight components that had to print reliably on their Bambu Lab H2S printer.

WHAT I DID

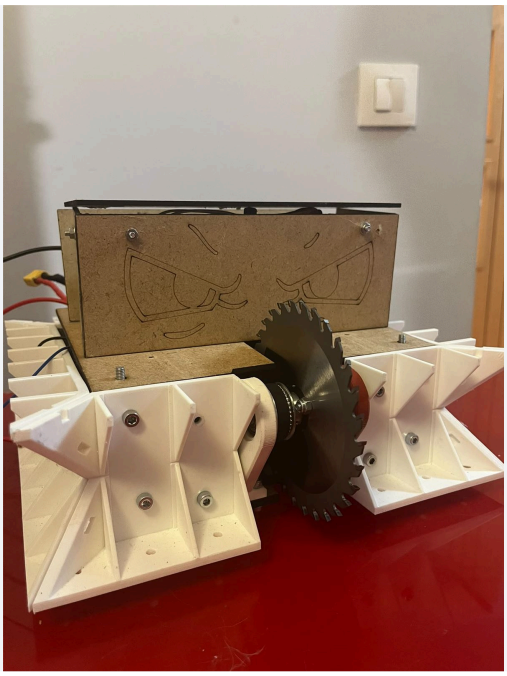
- Designed stackable modules with four angled plant pots and a fit between levels.
- Designed a conical water tank with a bamboo texture for the base.
- Added a mount for the Wi-Fi watering controller and a spacer for the irrigation pump.
- Optimised every part for FDM: orientation, supports, tolerances and splitting large parts into printable pieces.

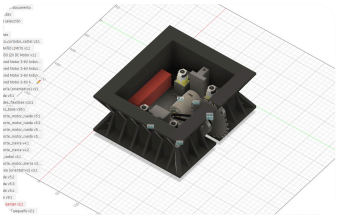
RESULT

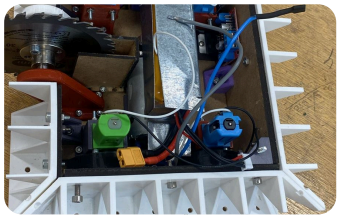
> The tower was printed and assembled. The client gave a **5-star review** — delivered on time and on budget.

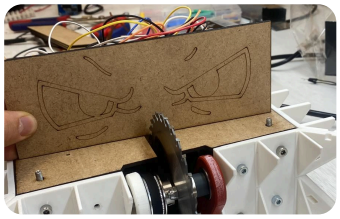
TOOLS

Fusion 360 · Bambu Studio · STL / 3MF









UNIVERSITY TEAM PROJECT

Panzer · Combat Robot

2026 · My role: mechanical design lead (Fusion 360) + firmware rework

CHALLENGE

Build a combat robot with a spinning saw-blade weapon for a university competition. It had to fit inside a maximum size, survive impacts and be easy to repair between fights.

WHAT I DID

- Designed the full robot in Fusion 360: more than 25 parts, all inside the size limit.
- Weapon: saw blade driven by an RS550 motor through a belt, with a custom shaft and supports.
- Drive: four geared motors with 3D-printed mounts, and flexible side walls to absorb hits.
- Chassis: laser-cut MDF combined with bolted 3D-printed parts.

RESULT

In the competition we **won one fight and drew the other**. A firmware bug and an electrical fault stopped the weapon from working properly.

WHAT I LEARNED — AND FIXED

> Over the summer I rewrote the firmware myself (ESP32, PS4 controller over Bluetooth, modular C++). **It now works reliably.** Next step: a **custom PCB** to replace the loose wiring and save space.





UNIVERSITY TEAM PROJECT

Underwater ROV

2026 · My role: mechanical design (Fusion 360) + motor-control firmware · Budget: under €500

CHALLENGE

Build a remotely operated underwater vehicle that can move in three axes and record video — on a student budget of under €500, a fraction of the price of a commercial ROV.

WHAT I DID

- PVC pipe frame joined with clamps and brackets that I designed and 3D-printed.
- Four thrusters: two vertical (up/down) and two horizontal (forward and turning).
- Wrote the motor-control firmware: ESP32 with H-bridge drivers and PWM speed control.
- Safety stop: all motors stop if the signal is lost for 3 seconds.
- Action camera and LED light mounted on the frame.

TOOLS

Fusion 360 · FDM printing · ESP32 / Arduino C++

// INDUSTRIAL EXPERIENCE

Fixing a production line

Warehouse Operator · Soft-drink bottling plant

Summer 2026 · Spain

Warehouse and forklift operation in a working production plant. When the date-coding printer on the filling line failed, I helped repair it:

1

Problem

The Hitachi RX-S inkjet coder stopped printing expiry dates. Cleaning the nozzle did not fix it.

2

Diagnosis

I read the English service manual and found the cause: a clogged filter plate inside the print head.

3

Fix

With my supervisor we disassembled the unit, cleaned the filter and reassembled it.

4

Result

The filling line restarted the same day, with no external technician call-out.

// SKILLS

What I work with

CAD & design

Fusion 360

SolidWorks

AutoCAD

Revit

Manufacturing

FDM 3D printing

Tolerances and fits

Laser cutting

STL · STEP · 3MF

Electronics & code

ESP32 / Arduino (C++)

Motor control (PWM)

PCB design (learning)

Analysis

MATLAB & Simulink

MathWorks Onramp certificate

Spanish Native

Ukrainian Native

English B1

German A1 · learning

// CONTACT

Let's build something.

Open to remote freelance projects and engineering internships.

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